Priyanshu Soni

Expertise in Numerical Modeling & Simulation (Fluid and Solid Mechanics)

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Professional Summary

CFD/Multiphysics researcher with four years of hands-on experience in fluid–structure interaction (FSI), pulsatile flow modelling, turbulence modelling, and experimental validation (PIV). Proficient in ANSYS Fluent, OpenFOAM, and COMSOL with strong programming skills in Python and MATLAB. Published in high-impact journals; PhD completion expected (dates of final seminar) July–November 2026.

Core Competencies

- CFD & FSI Simulation
- ANSYS Fluent / OpenFOAM / COMSOL
- Turbulence Modelling
- Reduced-Order Modelling
- PIV & Experimental Validation
- Python / MATLAB / ParaView
- CT/MRI Image Segmentation
- SolidWorks / Fusion 360 / AutoCAD
- Topology & Design Optimisation

Professional Experience

Doctoral Researcher — Computational Biomechanics *IIT BHU, Varanasi* 2023–Present

- Developed patient-specific high-fidelity CFD models of pulsatile arterial and ventricular blood flow from clinical CT/MRI datasets using numerical techniques.
- Performed coupled FSI simulations incorporating arterial wall viscoelasticity for multi-pathology vascular environments (stenosis–aneurysm interactions).
- Designed and validated experimental flow loop; conducted PIV measurements for numerical benchmarking.
- Developed reduced-order models, Physics informed neural network framework for reducing computational cost by significant margins while maintaining accuracy.
- Quantified hemodynamic indices (WSS, OSI, RRT, ECAP, HOLMES) using Python and MATLAB.
- Arterial phantoms for flow visualization via additive manufacturing and lost-core casting.

Junior Research Fellow (ICMR Project) *IIT BHU, Varanasi* Jul–Dec 2022

- Contributed to a government-funded computational cardiac model for non-invasive prediction of heart failure (Grant: INR 52,31,900).

Design Intern *Mechaton Engineering Pvt. Ltd* Jun–Aug 2021

- Design optimisation of multi-plunger positive displacement pumps using SolidWorks and ANSYS.

Testing Engineering Intern *National Engineering Industries (NEI) Ltd, Jaipur* Mar–Jul 2018

- Testing of precision inner-track/bore grinding machines; bearing track roundness analysis in metrology lab.

Selected Publications (6 peer-reviewed journals, 1 book chapter)

- Soni, P. *et al.* (2025). Pulsatile flow dynamics in an artery with multiple pathologies: An FSI study. *Physics of Fluids*, 37(7), 071913.
- Soni, P. *et al.* (2024). Comprehensive review on CFD simulations of left ventricle hemodynamics. *J. Brazilian Soc. Mech. Sci. & Eng.*, 46, 301.
- Soni, P. *et al.* (2026). Numerical modelling of pulsatile blood flow in 3D left ventricle models. *J. Flow Visualization & Image Processing* (Accepted).
- Soni, P. *et al.* (2026). Structural and functional optimisation of bio-ceramic coatings on topology-optimised mandibular implant. *J. Australian Ceramic Society.*

Education

PhD — Cardiovascular Biomechanics	IIT BHU Varanasi	*Present, CGPA 9.19*
M.Tech — Design Engineering (Mechanical)	NIT Hamirpur	*2022, CGPA 8.73*
B.Tech — Mechanical Engineering	RTU (Govt. Eng. College, Ajmer)	*2019, 63.6*

Awards, Grants & Certifications

- **Travel Grant** — Society of Mathematical Biology (SMB), ACMB–JSMB 2025, Kyoto, Japan
- **PhD Research Fellowship** — Government of India
- **ANSYS Analysis Engineer Training** (Udemy) — Static, Thermal, Dynamic & Topology Optimisation
- **SolidWorks 2021 Essential Training** — LinkedIn Learning

Languages

English (Professional) — Hindi (Native)